

PROSTHETIC SOCK

Project90





7,700+
CASES

The background of the image shows several people sitting on chairs, using crutches. The image is overlaid with a semi-transparent blue filter. In the center, there is a large circle divided horizontally into two halves: a light blue top half and a dark blue bottom half. Text is centered within this circle.

**7,700+
CASES**

**SOCKET FIT
REMAINS A
PERSISTENT
CHALLENGE**



**THE FIT IS
NEVER
PERFECT**

The image features three overlapping circles on a blue background. The leftmost circle is dark blue and contains the text 'THE FIT IS NEVER PERFECT' in white. The top-right circle is a medium blue and contains the text 'HEALTH RISKS' in white. The bottom-right circle is a light blue and contains the text 'DELAYING REHAB' in white. The background is a solid blue color with a faint, darker blue pattern of vertical lines.

**THE FIT IS
NEVER
PERFECT**

HEALTH RISKS

**DELAYING
REHAB**

EXISTING SOLUTIONS



<https://shop.ottobock.ca/en/EasyScan>

Industry

Hand-held 3D scanners

**Performed without load or
socket**

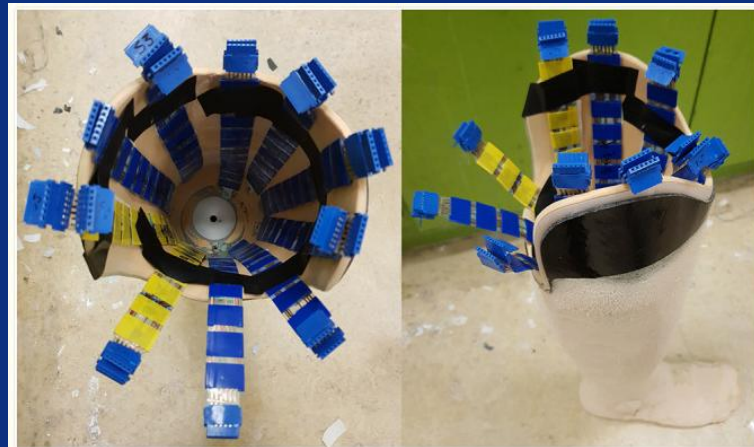
Expensive

EXISTING SOLUTIONS

Academic

Uniformly placed Force Sensitive Resistor (FSR) sensor strips

Lack of sensor coverage may lead to inaccurate data



<https://pmc.ncbi.nlm.nih.gov/articles/PMC9247243/>

ONGOING NEED

Socket fit needs can continue as a patient's limb and activity change over time

Fit matters during initial fitting and throughout prosthesis use

PROSTHETIC SMART SOCK

The smarter, safer way to fit prosthetics

THIS IS THE SMART SOCK



Easily available electronics
for data collection

Simple pressure
distribution map

Sensors optimally placed
for accuracy

ADDRESS THE WEAKNESSES

**Real time
mapping**

Static and Dynamic
positions

Cost-effective

Easily created and
affordable

**Resolution where
needed**

More targeted
sensor coverage

**Clinically
Meaningful**

Easy for practitioners
to use

TIMELINE



**Use easy-access
components for
prototype**



**Data collection
and finalizing
sensor locations**



**Testing on
multiple
materials**



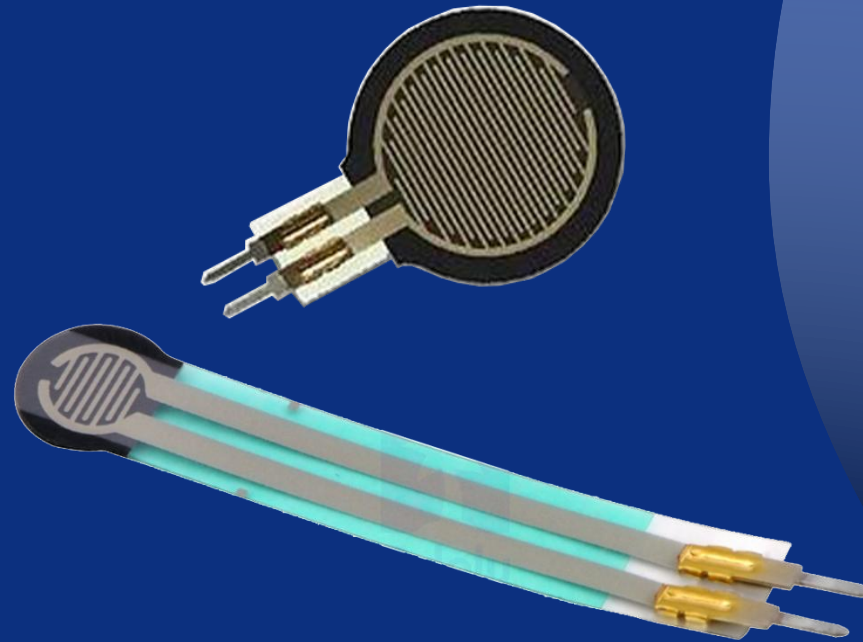
Final prototype



**Prepare for
controlled bench
testing**

SENSOR PROTOTYPE

- Finding commercial sensors for initial data
- Transferring data using conductive thread
- Testing load distribution



DATA & SAFETY

- Creating a visual map of pressure distribution
- Preventing long term wear of electronics
- Accessible and safe designing



FINAL PROTOTYPE

- Optimal force sensor placement in sock
- Durable and consistent force sensors
- Account for multiple material properties



PATIENT FEEDBACK

- Create low-training requirement GUI
- Make data relative (easier to interpret)
- Assess comfort of the sensor sock



COMMERCIAL PRODUCT

- Gain trust through clinics and reviews
- Start making sensors using textiles and PCBs
- Scale with custom more tailored manufacturing



REGULATORY CONSIDERATIONS

**Device
risk class**

**Confirm
licensing with
Health Canada**

**Ethics
approval
before human
studies**

**Confirm requirements before human studies or
commercialization**

INTELLECTUAL PROPERTY



UNIVERSITY OF
CALGARY

 **Cascade**
Prosthetic Services Ltd.

PROJECT

90

SCALABILITY



Calgary:

**Working with one
company**

Use local collaboration
to evaluate the fitting
workflow



SCALABILITY



Alberta:

Broader adoption
requires validation
across different
clinical workflows

Clinical
partnerships can
guide responsible
expansion

National expansion
is a long-term goal



LET'S MAKE COMFORT MORE ACCESSIBLE

Created by:

